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1. Stack and Queue.

Stack operations:

- push (54) -> [54]
- push (52) -> [54, 52]
- pop () -> removes 52.
- push (55) -> [54, 55]
- push (62) -> [54, 55, 62]
- s = pop() -> s = 62.

Queue operations:

- enqueue (21) -> [21]
- enqueue (24) -> [21, 24]
- dequeue () -> remove 21
- enqueue (28) -> [24, 28]
- enqueue (32) -> [24, 28, 32]
- q = dequeue () -> q = 24

Answers

s + q = 62 + 24 = 86.

Expression: 10 5 + 60 6 / * 8 -

Evalution:

- $10 + 5 = 15$
- $60 / 6 = 10.$
- $15 \times 10 = 150.$
- $150 - 8 = 142.$

Answers = 142

3. Gate CSE 2007

Postfix : 8 2 3 ^ / 2 3 * + 5 1 * -

After evaluting the first * :

- $2^3 = 8.$
- $8 / 8 = 1$
- $2 \times 3 = 6.$

stack becomes : [1, 6]

Answers : 6 and 1.

4. infix : a + b * c - d ^ e ^ f.

since ^ is right associative :

- $d^{(e^f)}$

Postfix : a b c * + d e f ^ ^ -

Answer : abc+def ^^-*

5. Gate CSE 1991.

operations:

* push (10)
* push (20)
* pop → (20)
* push (10)
* push (20)
* pop → 20
* pop → 10
* pop → 10
* push (20)
* pop → 20.

sequence: 20, 20, 10, 10, 20.

Answer: 20, 20, 10, 10, 20.